3.1.1 Atomic structure: Sheet 1

Isotopes and mass spectrometry · 21 marks · about 25 minutes

Key information

- Isotopes have the same number of protons but a different number of neutrons.
- $\mathbf{A_r}$ = Σ(m/z × abundance) ÷ total abundance, compared with 1/12 of the mass of a ^{12}C atom.
- Electron impact: $X(g) + e^- \rightarrow X^+(g) + 2e^-$. Electrospray: $X(g) + H^+ \rightarrow XH^+(g)$.
- All ions get the same kinetic energy, so lighter ions travel faster and reach the detector first.
- $\mathbf{KE = \frac{1}{2}mv^2}$ and $\mathbf{v = d \div t}$. Work in kg and m; mass of one ion (kg) × 1000 → g; × L → mass of one mole.
- Ions gain an electron at the detector, creating a current; abundance ∝ size of the current.

Section A: Self-quiz

01 Give the meaning of the term relative atomic mass.

[2 marks]

02 State the meaning of the term electron impact ionisation.

[1 mark]

03 A sample of germanium is analysed in a TOF mass spectrometer using electron impact ionisation. Give an equation, including state symbols, for the process that occurs during the ionisation of a germanium atom.

[1 mark]

04 Which atom contains the most neutrons?

[1 mark]

A ^{54}Cr
B ^{55}Mn
C ^{57}Fe
D ^{58}Ni

05 In a time of flight mass spectrometer, molecule X is ionised using electrospray ionisation. What is the equation for this ionisation?

[1 mark]

A $X(l) + e^- \rightarrow X^+(g) + 2e^-$
B $X(g) + e^- \rightarrow X^+(g) + 2e^-$
C $X(l) + H^+ \rightarrow XH^+(g)$
D $X(g) + H^+ \rightarrow XH^+(g)$

Section B: Exam questions

06 A TOF mass spectrometer is used to determine the relative atomic mass of a metal.

06.1 Explain why the sample of the metal must be ionised before passing into the flight tube. Describe how the ion is detected. Describe how the abundance of each isotope is determined.

[3 marks]

06.2 The mass spectrum showed four peaks. Table 1 shows data about the four peaks in this spectrum.

Table 1

m/z	Percentage abundance
50	4.3
52	82.8

m/z	Percentage abundance
53	10.5
54	2.4

State the m/z value of the ion that would be the last to reach the detector.

[1 mark]

06.3 Calculate the relative atomic mass of this sample of metal.

Give your answer to 1 decimal place.

[2 marks]

06.4 These are the data for one of the isotopes in this analysis.

Length of flight tube = 0.850 m Time of flight = 6.41×10^{-4} s KE = 7.59×10^{-20} J

The kinetic energy of an ion is given by the equation $KE = \frac{1}{2}mv^2$ (KE = kinetic energy / J, m = mass / kg, v = speed / m s^{-1})

Calculate the mass, in g, of one ion of this isotope.

[3 marks]

07 A sample of chromium contains four isotopes and has a relative atomic mass of 52.09

Table 2 shows the mass number and the percentage abundance of three of these isotopes.

Table 2

Mass number	52	53	54
Abundance (%)	82.8	10.9	2.7

Determine the percentage abundance of the fourth isotope.

Show by calculation that the mass number of this isotope is 50

[3 marks]

08 Time of flight (TOF) mass spectrometry is an important analytical technique.

A mixture of three compounds is analysed using a TOF mass spectrometer. The mixture is ionised using electrospray ionisation.

The three compounds are known to have the molecular formulas $C_3H_5O_2N$, $C_3H_7O_3N$ and $C_3H_7O_2NS$

08.1 Describe how the molecules are ionised using electrospray ionisation.

[3 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	M1 Average / mean mass of 1 atom (of an element) M2 ÷ 1/12 mass of one atom of ^{12}C OR (Weighted) average mass of all the isotopes ÷ 1/12 mass of one atom of ^{12}C	Source: 7404/1 Jun 2025 Q03.3 Mark top and bottom line independently. All key terms must be present for each mark. If moles and atoms mixed, max = 1	2
02	(High energy) electrons (from an electron gun) are used to knock out an electron (from each molecule or atom)	Source: 7404/1 Jun 2023 Q07.2	1
03	$Ge(g) + e^- \rightarrow Ge^+(g) + 2e^-$ OR $Ge(g) \rightarrow Ge^+(g) + e^-$	Source: 7404/1 Jun 2019 Q02.3 State symbols essential	1
04	**C**	Source: 7404/1 Jun 2023 Q09 ^{57}Fe (31 neutrons)	1
05	**D**	Source: 7404/1 Nov 2021 Q11	1
06.1	M1 So it can be accelerated / so it can be detected M2 Ion (hits negative plate or detector to) gain an electron M3 Abundance ∝ current flow / idea that abundance is proportional to current	Source: 7404/1 Jun 2025 Q03.4–03.7 M2 allow ion knocks out an electron into electron multiplier M3 allow signal from electron multiplier proportional to number of ions	3
06.2	54		1
06.3	M1 [(50 × 4.3) + (52 × 82.8) + (53 × 10.5) + (54 × 2.4)] ÷ 100 M2 52.1	Must be 1 dp	2
06.4	M1 $v = d \div t = 0.850 \div 6.41 \times 10^{-4} = 1326$ (m s^{-1}) M2 $m = 2KE \div v^2 = (2 \times 7.59 \times 10^{-20}) \div 1326^2 = 8.63 \times 10^{-26}$ (kg) M3 m = M2 × 1000 = 8.63×10^{-23} (g)	M2 uses their M1; M3 uses their M2 Alternative: M1 $m = 2KEd^2 \div t^2$, then M2 and M3 as above	3
07	M1 % of 4th isotope = 3.6 M2 [(52 × 82.8) + (53 × 10.9) + (54 × 2.7) + 3.6x] ÷ 100 = 52.09 M3 179.9 = 3.6x and x = 49.97 = 50 (evidence of working)	Source: 7404/1 Jun 2019 Q03.3 Allow alternative methods, eg M2 (52 × 82.8) + (53 × 10.9) + (54 × 2.7) + (50 × 3.6) = 5209 and M3 A_r = 5209 ÷ 100 = 52.09 M3 awarded for 50x = 179.9 and then x = 3.6	3
08.1	M1 (Sample is) dissolved (in a volatile solvent) M2 (Injected through) needle / nozzle / capillary at high voltage / positively charged M3 Each molecule / particle gains a proton / H^+	Source: 7404/1 Jun 2019 Q02.1 Allow named solvent (eg water / methanol) Ignore pressure Allow M3 from a suitable equation; do not allow atoms gain a proton for M3 Ignore references to electron gun ionisation	3